

Mathematica in Word Pattern Avoidance Research

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Abstract. We discuss our long-lasting extensive research from 1990 to the present on abelian pattern avoidance in words and the use of *Mathematica* as a programming tool and computing environment which has been crucial in interactive development of the code, in visualizations, and in extensive distributed computations. *Mathematica* has enabled us to discover phenomena that otherwise would have been inaccessible or would have been regarded as unbelievable.

Our recent research findings include a powerful abelian square-free substitution over four letters. The key characteristics of the new substitution derive from delicate mutations in the image words that merely emerged from the computations. We do not think it would have been feasible to create them by any design. Another quite recent finding concerns unfavourable factors that can be used to explain, at least partly, both the highly non-linear behavior in our earlier computations and the extreme difficulty that has dominated the search for abelian square-free endomorphisms and substitutions over four letters.

This paper contains a number of visualizations of the structures and processes.

1 Introduction

The systematic study of word structures, i.e., combinatorics on words, was started by Axel Thue (1863–1922) in [37] at the beginning of the 20th century. One of his discoveries was that consecutive repetitions of non-empty factors (squares) can be avoided in infinite words over a three-letter alphabet. As a simple example of the square concept, consider the words *abacaba* and *ab cd cd ab*. The first word does not contain any square, i.e., it is square-free, whereas the second word contains the underlined square *cd cd* as a factor.

The above-mentioned square-freeness property of words is not trivial to prove. The tool which Thue invented for constructing square-free, and other repetition-free, words, namely the concept of a repetition-free morphism, is still a basic technique in the study

of avoidable patterns in words. Repetition-free morphisms are mappings between free monoids that preserve the repetition-freeness of words. The iteration of a non-trivial repetition-free endomorphism or substitution (that maps a letter to more than one word) produces repetition-free words of any length. Dealing with substitutions somewhat later, we point out that repetition-free morphisms have been sharply characterized in [5, 10, 19, 20, 27, 28, 34, 38]. The results therein concern different types of repetitions (k -repetitions for a given integer $k \geq 2$) and alphabet sizes. Informally speaking, most of the characterizations mean that it is possible to test the repetition-freeness of a given morphism just by checking whether the image words of short repetition-free words are also repetition-free. A general survey of these and related results, achieved before 1984, is given in [3]. For a short survey of Thue's results concerning repetition-free words and their applications, see [16]. Fundamental topics are discussed in [29, 36].

In a paper from 1961, see [15, p. 240], Paul Erdős (1913–1996) raised the question whether abelian squares can be avoided in infinitely long words, i.e., whether there exist infinitely many abelian square-free words over a given alphabet. Here, an abelian square means a non-empty word uv , where u and v are permutations (anagrams) of each other. For example, abc acb is an abelian square. A word is called abelian square-free, if it does not contain any abelian square as a factor. For example, the word abacaba is abelian square-free, while ab cabdc bcacd ac is not.

Later, in a 1970 paper, Pleasants [33] showed that there exists an infinite abelian square-free word over five letters. Finally, in 1991, see Keränen [21] the year after, we managed to show that the same holds true also in the case of four letters. It is easily seen that abelian squares cannot be avoided over a three-letter alphabet. Indeed, in this alphabet, each word of length 8 contains an abelian square. In [14] Entringer et al. showed that every infinite word over a binary alphabet contains arbitrarily long abelian squares. Dekking [12] in turn proved that abelian repetitions to the fourth power can be avoided in infinite words over two letters, and abelian repetitions to the third power (cubes) can be avoided in infinite words over three letters. For a generalization of abelian squares, see Avgustinovich and Frid [2]. Abelian fractional powers were studied by Cassaigne and Currie [8]. In [11], Currie showed that the number of binary words avoiding abelian fourth powers grows exponentially, and in [1], Aberkane, Currie, and Rampersad showed that the number of ternary words avoiding abelian cubes grows exponentially as well.

An application of Dekking's result was given by Justin et al. in [17], where it was shown that a finitely-generated semigroup is uniformly repetitive if and only if it is finite. In [32], Pirillo et al. used similar reasoning when proving, among other results, that the additive semigroup $\mathbb{N}^+$ is not uniformly 4-repetitive. It seems to be an open problem whether $\mathbb{N}^+$ is uniformly 2-repetitive or 3-repetitive. In all these considerations the use of van der Waerden's theorem has been very central. In Lothaire [29, pp. 55–62] van der Waerden's theorem was used to show that every morphism from a free semigroup A^+ , where A is finite, to $\mathbb{N}^+$ is repetitive. This means that every long enough sequence on a finite set of integers contains two adjacent segments (not necessarily of the same length) that have the same sum.

The original problem of abelian squares has also attracted attention in the study of free partially-commutative monoids, see for instance [9, 13]. Moreover, abelian square-free words have aroused interest in algorithmic music (Laakso [26]) and quite

recently in cryptography (Rivest [35], Bouillaguet et al. [4]).

In 1993, Carpi [5] gave sufficient conditions for morphisms to preserve abelian k th power-freeness of words. A conjecture is that these conditions yield an effective characterization also for abelian square-free endomorphisms on a four-letter alphabet $\Sigma_4 = \{a, b, c, d\}$. However, new examples of relatively short abelian square-free endomorphisms g of Σ_4^* have turned out to be extremely hard to find – and the same difficulty applies to every systematic attempt for constructing long abelian square-free words over 4 letters. Before our current findings, we were not at all optimistic that it would be possible to find more examples of abelian square-free endomorphisms – not to speak of proper substitutions of Σ_4^* . Thus far, since 1992 when we presented g_{85} in [21], the only new abelian square-free endomorphisms and substitutions have been found by Carpi [6], cf. also [7, pp. 80–81]. However, his mappings are all based on the structure of g_{85} . Moreover, the size of these endomorphisms and substitutions are large. By using these substitutions, Carpi showed that the number of abelian square-free words of each length grows exponentially, and that the monoid of (uniform) abelian square-free endomorphisms of Σ_4^* is not finitely generated.

Very recently, we succeeded in finding 200 new abelian square-free endomorphisms and some of them work as a starting point for a new powerful abelian square-free substitution. These endomorphisms have the property that the image words $g(x)$, $x \in \Sigma_4$ are all obtained by cyclically permutating the letters in $g(a)$. The image words $g(a)$ can be viewed and copied from the Internet [18]. The same cyclic permutation property is true for g_{85} as well, and this method was already used by Pleasants [33] in connection with five letters. Consequently, all of these endomorphisms have a uniform modulus and the generated words grow uniformly. The size of Pleasants' endomorphism is $5 \times 15 = 75$. In our case, we have checked using computers that the size $4 \times 85 = 340$ for g_{85} , in spite of its largeness, is actually minimal, at least as far as cyclic permutation method is used. So far, the search for other kinds of abelian square-free endomorphisms of Σ_4^* has not been very successful, even with extensive experimentation. However, we show later in (1) that 20724 (out of 20736) abelian square-free endomorphisms indeed possess a different structure.

Moreover, in 2002 we [22, 23] found a nice endomorphism g_{98} of Σ_4^* that can be used in iterations, and together with g_{85} to produce infinite abelian square-free DTOL-languages (i.e., languages obtained by using compositions of morphisms). This g_{98} itself is not an abelian square-free endomorphism, as it does not preserve abelian square-freeness for all words (starting with already from the length 7). The structure of the image words of g_{98} also partly differs from that of the other above-mentioned 201 (remembering also g_{85}) cyclic endomorphisms.

Quite recently, we have also gained insight as to why these abelian square-free structures are so rare. In [24] we explain, at least partly, this rareness of long words avoiding abelian squares by using the concept of unfavourable factor. We take an abelian square-free word and, using *Mathematica*, try to extend it in abelian square-free fashion to the right and to the left in all possible ways up to a given upper bound for the total length. At a time, increasing the length of the word by a given fixed length at each step. We extend alternately to right and left, and backtrack if necessary. If the given upper bounds are reached then the original word is a *so-far-favourable* one (it may still turn out to be unfavourable on later experiments). If there is no way to

reach the upper bounds, then the original word is classified, without any doubt, to be as unfavourable. Thus we obtain three kinds of words: *unfavourable* (*bad*), *so-far-favourable* (*so-far-so-good*), and *favourable* (*good*).

It is a remarkable phenomenon that sometimes relatively short so-far-favourable words turn out to be unfavourable factors after being ‘safely’ extendable (to the right and left) for quite a long distance and with a really huge number of branches. One might have expected the quite long buffers to guarantee the further growth. We suspect that the majority of abelian square-free words over four letters cannot occur as proper factors in the middle of very long (infinite) abelian square-free words. In a way, the experimental facts concerning unfavourable factors explain the highly non-linear behavior of our earlier computations and also the difficulty of finding abelian square-free endomorphisms of Σ_4^* (not to speak of substitutions). At present we know that in the four letter case about 60 % of the abelian square-free words of length 24 are indeed unfavourable.

It will be interesting to study, in a similar way, the case of three letters, for which an exciting open problem was posed by Mäkelä [30], who allows repetitions xx and xxx , for a letter x , but no other abelian squares (or cubes).

2 Preliminaries for combinatorics on words

In this section we present notation and terminology. Our terminology is quite standard in the field of combinatorics on words. Consequently, the reader may consult this section later, as needed.

An *alphabet* Σ is a finite non-empty set of abstract symbols called *letters*. A *word* (*string*) over Σ is a finite (unless otherwise indicated) string, or sequence, of letters belonging to Σ . The set of all words over Σ is denoted by Σ^* , while the set of non-empty words is denoted by Σ^+ . For words u and v in Σ^* , the associative binary operation of *catenation* is defined as the juxtaposition uv . The *empty word*, which is the neutral element of catenation, is denoted by λ . The algebraic structures Σ^* and Σ^+ are called, respectively, the free monoid and the free semigroup generated by Σ .

Let $w = x_1 \cdots x_m$, $x_i \in \Sigma$. The *length* of the word w , denoted by $|w|$, is the number of occurrences of letters in w , i.e., $|w| = m$. Let $\Sigma = \{a_1, \dots, a_n\}$. The number of occurrences of one letter $x \in \Sigma$ in the word w is denoted by $|w|_x$, or simply by $|w|_i$ if $x = a_i$. The notation $\psi_\Sigma(w)$ stands for the *Parikh vector* of w , i.e., $\psi_\Sigma(w) = (|w|_1, \dots, |w|_n)$. Usually we will omit the subscript Σ and write simply ψ instead of ψ_Σ .

A word u is called a *factor* of a word w , if $w = p u s$ for some words p and s . The notation $\text{FACT}(w)$ stands for the set of all factors of w . If p (or s) = λ , then u is called a *prefix* (or a *suffix*) of w .

Let $k \geq 2$ be a integer. A *k-repetition* is a non-empty word of the form R^k . An abelian *k-repetition* is a non-empty word of the form $P_1 \cdots P_k$, where $\psi(P_\mu) = \psi(P_\nu)$ for all $1 \leq \mu < \nu \leq k$, i.e., P_i :s are *commutatively equivalent*, that is, they are permutations, or anagrams, of each other. Instead of [abelian] 2- and 3-repetitions, the terms [abelian] *squares* and *cubes* are often used. A word is called *k-repetition free*, or *k-free* for short, if it does not contain any *k-repetition* as a factor. A word sequence or a word set is *k-free*, if all words in it are *k-free*. Abelian analogs of these terms

and definitions also exist and are formed in a natural way by preceding any term with the word abelian, i.e., abelian square, abelian cube, abelian k -repetition free, etc. The abelian analog of the short term, k -free is a - k -free. If, for a fixed k , it is possible to construct arbitrarily long (infinite) a - k -free (or other pattern-free) words over a given alphabet Σ , then we say that abelian k -repetitions (or those patterns) are *avoidable* over Σ .

A *morphism* h is a mapping between free monoids Σ^* and Δ^* with $h(uv) = h(u)h(v)$ for every u and v in Σ^* . In particular, $h(\lambda) = \lambda$. A morphism $h : \Sigma^* \rightarrow \Delta^*$, being compatible with the catenation of words, is uniquely defined, if the word $h(x) \in \Delta^*$ is (effectively) given for each $x \in \Sigma$. If $\Delta = \Sigma$, we call h an *endomorphism* (and usually write g instead of h). For a morphism h and a language L we define $h(L) = \{h(w) | w \in L\}$. A morphism h is called *uniformly growing*, or is said to have a *uniform modulus*, if $|h(x)| = |h(y)| \geq 2$ for every x and $y \in \Sigma$.

A *substitution* $\sigma : \Sigma^* \rightarrow 2^{\Delta^*}$ is a monoid morphism of Σ^* into a subset monoid of Δ^* . The substitution σ can be regarded as a multi-valued mapping between the free monoids Σ^* and Δ^* , and written $\sigma : \Sigma^* \rightarrow \Delta^*$. The substitution σ is finite if $\sigma(\Sigma)$ is a finite subset of Δ^* . Obviously, for a morphism $h : \Sigma^* \rightarrow \Delta^*$, it holds that $\text{Card}(h(\Sigma)) \leq \text{Card}(\Sigma)$, and thus a morphism is a special case of a finite substitution. Following the terminology of Carpi [7], a substitution $\sigma : \Sigma^* \rightarrow \Delta^*$ is called *commutatively functional*, if $\text{dom}(\sigma) = \Sigma^*$, and, for all $x \in \Sigma$, $v' \in \sigma(x)$, it holds that $\psi(v) = \psi(v')$ (this is also written as $v \sim v'$). In other words, a substitution is termed commutatively functional, if the image words of a fixed letter are all commutatively equivalent. Moreover, for a commutatively functional substitution σ and any word w in Σ^* , all the words $\sigma(w)$ are commutatively equivalent.

For a given integer $k \geq 2$, a substitution (or a morphism) $\sigma : \Sigma^* \rightarrow \Delta^*$ is called *k -free* [*a - k -free*], if all the words $\sigma(w)$ are (or the word $\sigma(w)$ is) k -free [a - k -free] for every k -free [a - k -free] word $w \in \Sigma^*$.

3 The new a -2-free substitution σ_{109} over 4 letters

Let $\Sigma_4 = \{a, b, c, d\}$. Define the substitution $\sigma_{109} : \Sigma_4^* \rightarrow \Sigma_4^*$ as follows. First let the 12 image words of $\sigma_{109}(a)$, say $\{A_1, A_2, \dots, A_{12}\}$, have the form

$$A_i = p_{16} w_4 u_{27} w_3 s_{59} = abcacdcbedcadcdcb w_4 badacdadbdcdbdabdbcbabcbdc b w_3 bdcdadcdcbabcbdc bca cdcacbadabcbdc bca dbabcbabdbcbdbadbdcbca,$$

with 12 different factor pairs (w_4, w_3) , taken in the natural lexicographical order form $\{abcd, abdc, adbc, dacb\} \times \{acd, adc, cad\}$. The subscripts of the factors $p_{16}, w_4, u_{27}, w_3, s_{59}$ indicate their lengths. Note that all the words in $\{abcd, abdc, adbc, dacb\}$, and respectively in $\{acd, adc, cad\}$, are commutatively equivalent. The delicate mutations in words of $\sigma_{109}(a)$ can also be described and visualized as a movement of letters d and c in the invariant background:

$$\begin{array}{llll} \text{dabc} & \text{cad} & , & \text{abdc} & \text{cad} \\ \text{dabc} & \text{acd} & , & \text{abdc} & \text{acd} \\ \text{dabc} & \text{adc} & , & \text{abdc} & \text{adc} \\ \text{adbc} & \text{cad} & , & \text{abcd} & \text{cad} \\ \text{adbc} & \text{acd} & , & \text{abcd} & \text{acd} \\ \text{adbc} & \text{adc} & , & \text{abcd} & \text{adc} \end{array}.$$

To complete the definition of σ_{109} , let $\sigma_{109}(\phi(x)) = \phi(\sigma_{109}(x))$ for all $x \in \{a, b, c, d\}$, where $\phi : \Sigma_4^* \rightarrow \Sigma_4^*$ is the circular letter-to-letter endomorphism defined by $\phi(a) = b, \phi(b) = c, \phi(c) = d, \phi(d) = a$. Thus, informally, the set of image words for b, c, d are obtained, letter-by-letter, by cyclic permutation of letters of all the words in $\{A_1, A_2, \dots, A_{12}\}$. Obviously, σ_{109} is a commutatively functional substitution of Σ_4^* (of uniform modulus 109). The Parikh vectors for the image words of letters are the rows of the matrix below:

$$\begin{pmatrix} \psi(A) \\ \psi(B) \\ \psi(C) \\ \psi(D) \end{pmatrix} = \begin{pmatrix} 21 & 31 & 29 & 28 \\ 28 & 21 & 31 & 29 \\ 29 & 28 & 21 & 31 \\ 31 & 29 & 28 & 21 \end{pmatrix},$$

whenever

$$A \in \sigma_{109}(a), B \in \sigma_{109}(b), C \in \sigma_{109}(c), D \in \sigma_{109}(d).$$

Using a computer, we checked the a-2-freeness of σ_{109} in two (albeit not completely) different ways. The first way was a direct but long method similar to what we used previously in the paper [21] in 1992. There, the code development was done in LISP. In the present work, we used *Mathematica* to make most of the computational steps visible, thus providing a way to recheck the result. In these computations we benefited greatly from *Mathematica*'s dynamic programming feature, which guarantees that functions remember the values they have found. The second method that we used for checking the a-2-freeness of σ_{109} , is an application of Carpi's [7] characterization. The details of that method are explained in [25] and lie outside the main topic of this paper, though, we may mention, in passing, that in this case as well, it was natural to develop the algorithms by using *Mathematica*.

In connection with the substitution σ_{109} , let us consider the $12^4 = 20736$ different endomorphisms $g_{109,ijkl}$ of Σ_4^* , defined by

$$\begin{aligned} g_{109,ijkl}(a) &= A_i, & g_{109,ijkl}(b) &= B_j = \phi(A_j), & g_{109,ijkl}(c) &= C_k = \phi(B_k), \\ g_{109,ijkl}(d) &= D_l = \phi(C_l), & i, j, k, l &= 1, \dots, 12. \end{aligned} \quad (1)$$

Our checking of each show that all 12^4 endomorphisms are indeed abelian square-free. This alone suggests that the substitution σ_{109} might really be a-2-free. Although additional tests are needed, they pose no particular difficulty. Indeed, in [25] we justify the following conclusion:

Proposition 1 *The substitution $\sigma_{109} : \Sigma_4^* \rightarrow \Sigma_4^*$ defined above is abelian square-free.*

Of course, the computational details should also be carried out independently of us. We hope that in the very near future people will accomplish this.

It is likely that new abelian square-free substitutions of Σ_4^* can be constructed not only from $g_{109,ijkl}$, but also from other a-2-free endomorphisms that we have recently found. The image word $g(a)$ for all 200 a-2-free endomorphisms, g_{85} (found in 1990), and g_{98} (found in 2002), can be viewed and copied from the Internet [18].

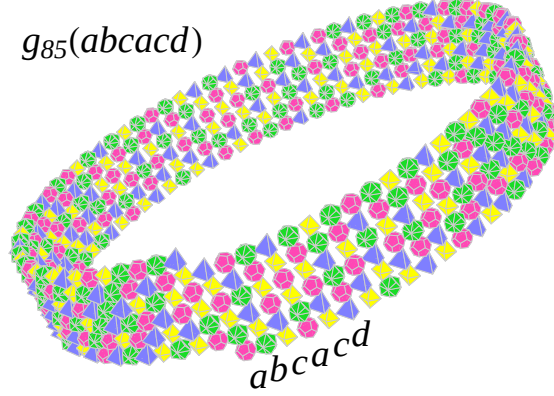


Figure 1: Six starting image words of a self-reading string for g_{85} .

The properties of σ_{109} lead to a considerably sharper lower bound for the exponential growth of c_n , i.e., of the number of a-2-free words over 4 letters of length n . We find that $c_n > \beta^{-50} \beta^n$ with $\beta = 12^{1/m} \simeq 1.02306$. For details the reader is referred to [25]. The exponential growth of c_n was first proved by Carpi [6], who showed that $c_n > \beta^{-t} \beta^n$ with $\beta = 2^{19/t} = 2^{19/(85^3-85)} \simeq 1.000021$, where $t = 85^3 - 85$ is the modulus of his substitution constructed from g_{85} that we presented in [21].

The number of all a-2-free words over 4 letters up to the length 60 can be found on the Internet [18].

4 Visualizations of structures and processes

Some of the visualizations presented in this section can also be found at our web pages. All of them are created using *Mathematica*. In many cases, the *Mathematica* graphics is imported into interactive tools, such as the LiveGraphics3D Java applet or applications developed by C++.

The pictorial representation of Figure 1 contains the six image words of the word $abcacd$ related to the iteration of the abelian square-free endomorphism g_{85} . In Figure 2, the visualization can be used to detect structures separately at even and at odd positions. In Figure 3, one finds the first twenty image words for the self-reading sequence associated with the endomorphism g_{98} of Σ_4^* . Note the pink diagonal consisting of occurrences of the letter a . In Figure 4, the word $(g_{98})^2(a)$ is represented by using only the letters in $\{b, d\}$ and leaving the occurrences of a and b white. In Figure 5, the directions for letters are indicated in a quad tree representation. Quad trees can be used to visualise large sets of words over 4 letters at a glance. In Figure 6, one sees how longer words can be represented by dividing the squares further. All abelian square-

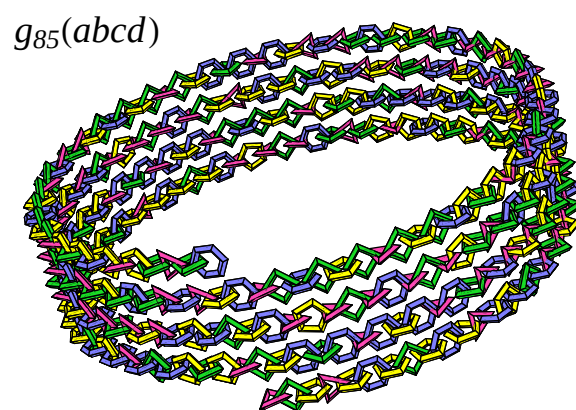


Figure 2: Even and odd positions separated.

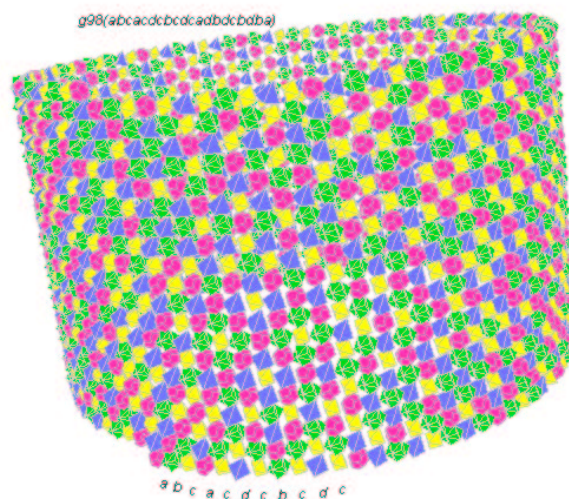


Figure 3: Twenty starting image words for g_{98} . Note the pink diagonal.

free words of length 2 over 4 letters are shown. Note the white corners representing the unfavourable words aa , bb , cc , dd . In Figure 7, the quad tree shows all the 3576 abelian square-free words of length 8 over 4 letters in one picture. In the Introduction we explained the concept of *unfavourable* (*bad*), *so-far-favourable* (*so-far-so-good*), and *favourable* (*good*) factors, and noted that it is a remarkable phenomenon that relatively short so-far-favourable words turn out to be unfavourable factors even being 'safely' extendable (to the right and left) for quite a long distance and sometimes with a really huge number of branches.

Most surprising in this respect is the behavior of $abcdacbabdabacdabcddad$. For this word of length 22, we obtain a list of pairs (x, y) , where x represents the length of the words in the so-far-favourable bi-directional tree, and y represents the number of all possible extensions of the length in question. The words, in the (somewhat abbreviated) list below, are extended by one letter at a time only, that is, all extensions of length 1 are tried.

$\{\{22, 1\}, \{23, 2\}, \{24, 2\}, \{25, 5\}, \{26, 14\}, \{27, 23\}, \{28, 14\}, \{29, 26\}, \{30, 10\}, \{31, 16\}, \{32, 8\}, \{33, 9\}, \{34, 9\}, \{35, 16\}, \{36, 16\}, \{37, 27\}, \{38, 27\}, \{39, 54\}, \{40, 54\}, \{41, 68\}, \{42, 136\}, \{43, 194\}, \{44, 291\}, \{45, 444\}, \{46, 296\}, \{47, 450\}, \{48, 225\}, \{49, 331\}, \{50, 331\}, \{51, 474\}, \{52, 948\}, \dots, \{107, 840479\}, \{108, 1679287\}, \{109, 2301836\}, \{110, 2302465\}, \{111, 3157227\}, \{112, 3154210\}, \{113, 4306159\}, \{114, 8466798\}, \{115, 11575001\}, \{116, 5779271\}, \{117, 7866918\}, \{118, 0\}\}.$

The death of all of the nearly 8 million branches of this bi-directional tree at length 117 looks dramatic. We remark that it was necessary to construct all the possible a-2-free words in the tree to be able to find the numbers and see the collapse. Consequently, this computation is quite a big, albeit a straightforward, one. Of course a much more massive search was needed to find this example in the first place. In the search for unfavourable factors, we have been using many features of *Mathematica*, including conversions from strings to symbols and to patterns, and further to cumulative integer lists. This part of the code uses a state-machine paradigm and the overall structure is actually quite complex. We do not think it would have been feasible for us to have developed the code and all the necessary pre-computational structures without the aid of *Mathematica*'s technical computing environment. In Figure 8, a part of the complex behavior of the above list for the unfavourable factor $abcdacbabdabacdabcddad$ is represented. In this Figure, we zoom into the middle behavior. The final peak of the list is omitted for clarity. Figure 9 is a visualization of $g_{85}(a)$ in the form of DNA. It would be desirable, if these kind of loops (hypohelix structures) were avoided in real DNA, since they may lead to diseases, see for example Mirkin [31]. It seems to be an open question to what extent the loops can be avoided over four letters (provided that the structure is not too trivial). This figure was designed by Erik Jensen, University of California, Berkeley, in 2005.

In Figures 10, 11, and 12, we represent 4-letter walks in the diamond lattice. The walks are obtained by first saving *Mathematica* graphics in a file and, then using Martin Kraus's LiveGraphics3D Java applet. The direction vectors for the letters a, b, c, d are shown in Figure 10. In Figure 11, the loops inside the walk represent factors containing an equal number of occurrences of each letter. The example word is the prefix $abcdcbcdcadcdabacababcbdbcb$ of $g_{85}(a)$.

Our last image, Figure 12, is obtained from interactive experiments with LiveGraphics3D. Rotating the figure of $g_{98}(a)$ and looking at it from the direction pointed

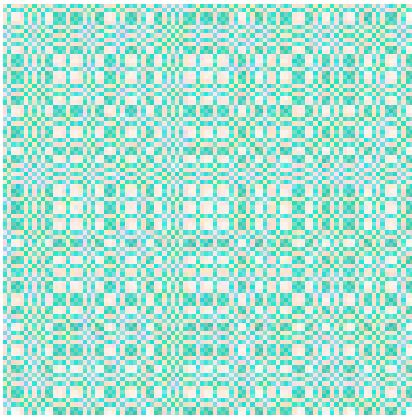


Figure 4: Occurrences of b and d in the word $(g_{98})^2(a)$.

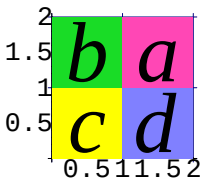


Figure 5: Directions for letters in a quad tree.

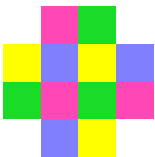


Figure 6: All abelian square-free words of length 2 over 4 letters.

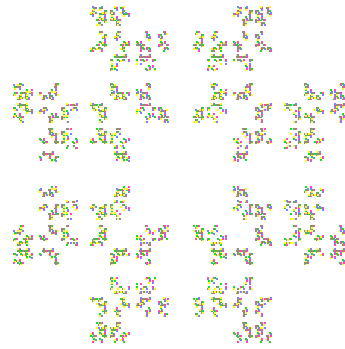


Figure 7: All abelian square-free words of length 8 over 4 letters.

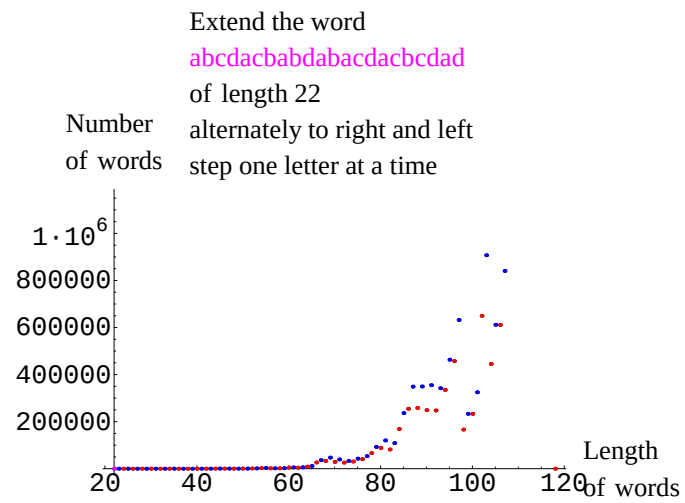


Figure 8: Unfavourable behavior of $abcdacbabdabacdabcddad$ when extended by one letter at a time.

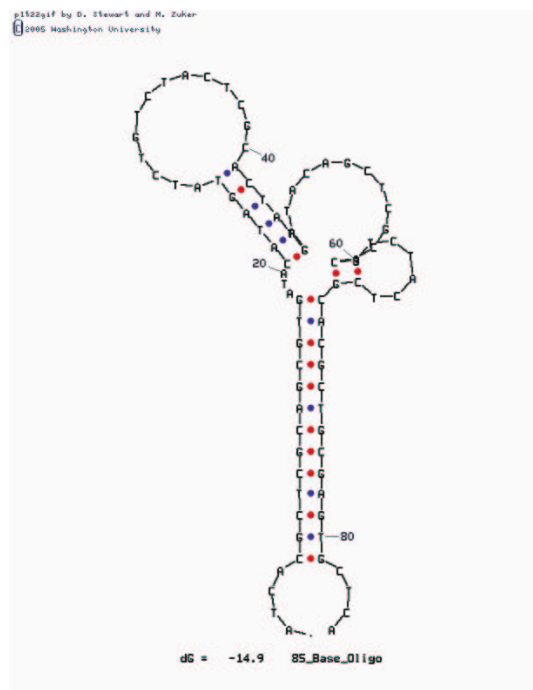


Figure 9: Hypohelix structures of $g_{85}(a)$.

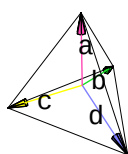


Figure 10: Direction vectors for letters a, b, c, d .

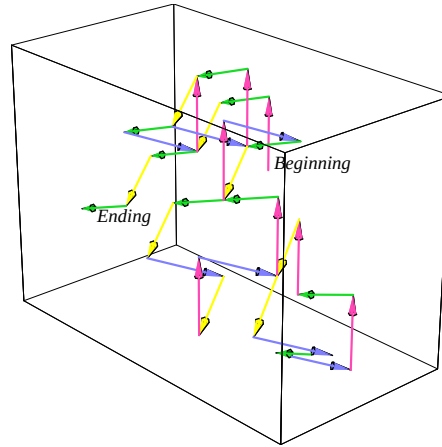


Figure 11: Loops represent factors containing an equal number of occurrences of each letter.

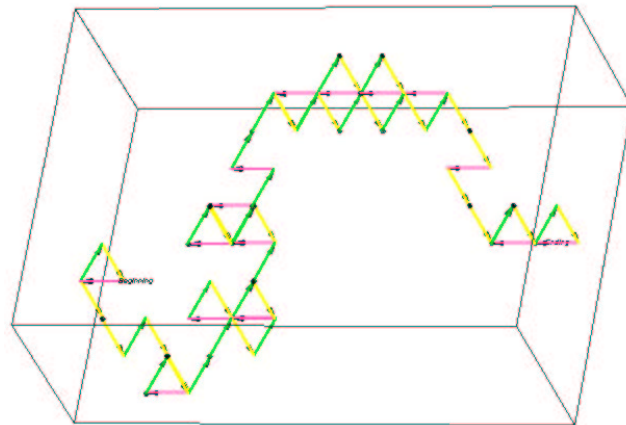


Figure 12: Semipalindrome structure of $dcdadbdcdbabdbcbacbcdabdc\ d\ bdcacd-bcbacbcdcacdbdcdadbd$ inside $g_{98}(a)$.

by the vector for d , one suddenly detects a semipalindrome structure of a long factor $dcdadbdcdbabdbcbacbcdabdc\ d\ bdcacdbcbacbcdacdcdbcdadbd$ inside this word.

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